

Experiment No. 7

Environment: Microsoft Windows

Tools/ Language: Oracle/SQL

Objective: To implement concept of various functions in SQL.

Theory & Concepts:

Oracle functions:

Functions make the basic query block more powerful and are used to manipulate data values. Functions accept one or more arguments and return one value. An argument is a user supplied constant, variable or column reference, which can be passed to a function in the following format:

function _name (argument1, argument2 ..)

Functions can be used to

- Perform calculations on data
- Modify individual data item
- Manipulate output for group of rows
- Alter date formats for display
- Convert column data types

Single Row Functions:

- Act on each row returned in a query
- Return one result per row
- Expect one or more user arguments
- May be nested

Single row functions can be of following types:

Character functions, Number Functions, Date functions and conversion functions.

ORACLE TABLE

DUAL: Dual table is owned by SYS. SYS owns the data dictionary; in DUAL is a part of data dictionary. Dual is a small Oracle worktable which consists of only one row and one column and contains the value *x* in that column. Besides arithmetic calculations, it also supports date retrieval and its formatting.

```
Select 3.14*3*3 from DUAL;  
28.26
```

```
Select 4/2 from DUAL;  
2
```

```
DESC DUAL;
```

Name	Null?	Type
DUMMY		VARCHAR2(1)

```
Select * from DUAL
```

D
--
X

```
Select sysdate from DUAL
SYSDATE
-----
17-FEB-14
```

- (a) **Character Functions:** accept character data as input and can return both character and number values.

LOWER (col/value)
UPPER (col/value)
INITCAP (col/value)
SUBSTR (col/value, position, n)
INSTR (col/value, 'string')
ASCII (character)
CHR (number)

- (b) **Number Functions:**

ROUND (col/val, n)
TRUNC (col/val, n)
CEIL (col/val)
FLOOR (col/val)
POWER (col/val, n)
SQRT (col/val)
EXP (n)
ABS (col/val)
MOD (value1, value2)

- (c) **Date Functions:** Date functions are used to manipulate and extract values from the date column of a table.

SYSDATE
ADD_MONTHS (date, count)
LAST_DAY (date)
MONTHS_BETWEEN (date2, date1)
NEXT_DAY (date, 'day')

- (d) **Conversion Functions:** Converts one data type into another.

TO_CHAR (input, format): Converts date or number into character string.
TO_DATE (date, format): Converts any date format to default format (dd-mon-yy).
TO_NUMBER (col/value): Converts a string into a number.

ORACLE FUNCTION EXAMPLES

NUMERIC FUNCTIONS

1. ROUND

```
Select ROUND(SUM(GPA)/COUNT(GPA)) as "Round Average"
from student
```

Round Average

4

Round(n , m) return rounded to m places to the right of a decimal point. If m is omitted then n is rounded to 0 places.

```
Select ROUND(15.81,1), round(158.158),round(158.158,-2) from
DUAL
```

```
ROUND(15.81,1)
-----
15.8
```

2. ABSOLUTE: This function return absolute value of numbers

```
Select ABS(-15) from DUAL
```

```
ABS(-15)
-----
15
```

3. POWER: Return m raise to n where n must be an integer.

```
Select POWER(7,3) from DUAL
```

```
POWER(7,3)
-----
343
```

4. SQUARE ROOT: Return the square root of n . If $n < 0$ then NULL.

```
Select SQRT(25) from DUAL
```

```
SQRT(25)
-----
5
```

```
select ROUND(sqrt(15),2) from DUAL
```

```
ROUND(SQRT(15),2)
-----
3.87
```

5. **EXPONENTIAL**

```
Select EXP(5) from DUAL  
EXP(5)
```

```
-----  
148.413159
```

6. **EXTRACT:** Returns a value extracted from a date or an interval. A date can be used only to extract year month or day while a time span with the time zone use to extract hours and minutes.

```
Select extract(month from sysdate)  
from DUAL
```

```
EXTRACT(MONTHFROMSYSDATE)  
-----  
2
```

7. **MOD:** Return the remainder of first number divided by second number passed as a parameter

```
Select MOD(15,7) from DUAL
```

```
MOD(15,7)  
-----  
1
```

8. **TRUNCATE:** Returns a truncated number to a certain number of decimal places
TRUNC(number, decimal places)

```
Select TRUNC(125.815,1) from dual
```

```
TRUNC(125.815,1)  
-----  
125.8
```

```
Select TRUNC(125.815,-2) from dual
```

```
TRUNC(125.815,-2)  
-----  
100
```

9. **FLOOR:** Returns the largest value that is equal to or less than the number.

```
Select floor(24.92), floor(3.1) from dual
```

```
FLOOR(24.92) FLOOR(3.1)  
-----  
24      3
```

10. **CEILING:** Return the largest value that is equal to or greater than the number.

```
Select ceil(24.92), ceil(3.1) from dual
```

```
CEIL(24.92) CEIL(3.1)  
-----  
25      4
```

STRING FUNCTIONS:

- 1. Lower:** Returns char, with letters in lower case.

```
Select LOWER('RAJeEv') from dual
```

```
LOWER(
----
Rajeev
```

- 2. UPPER:** Return char with letters in UPPER case

```
Select LOWER('rAJeEv')
from dual
```

```
LOWER(
----
RAJEEV
```

- 3. SUBSTRING:**

SUBSTR(string, start_pos, length)

```
Select SUBSTR('Prateek', 4, 3) from dual
```

```
SUB
---
tee
```

```
Select SUBSTR('Prateek', 4) from dual
```

```
SUB
---
teek
```

- 4. INSTR:** Returns the location of substring in the string

INSTR(string, string2, start position, nth appearance)

```
Select instr('Rahul Pradhan', 'a', 2, 3)
from dual
```

```
INSTR('RAHULPRADHAN','A',2,3)
-----
12
```

```
Select instr('Rahul Pradhan', 'a', 2) as INSTR
from dual
```

```
INSTR
-----
2
```

String: HTC launch a new phone on net

Search t.

```
Select instr('HTC launch a new phone on net', 't')
from dual
```

```
INSTR('HTCLAUNCHANEWPHONEONNET','T')
-----
29
```

5. TRANSLATE

```
Select translate('HTC launch a new phone on net', 'net',  
'web')as Translate from dual
```

TRANSLATE

HTC lauwh a wew phowe ow web

6. LENGTH

```
Select length('Pratyush Mehrotra') as Length from dual
```

LENGTH

17

7. TO_CHAR(n,fmt): Converts a number to a character

```
Select to_char(17475,'$099,999')as FMT from dual
```

FMT

\$017,475

```
Select TO_CHAR(sysdate, 'Month DD, YYYY') from dual
```

TO_CHAR(SYSDATE,'M

February 17, 2014

```
Select TO_CHAR(date '2014-02-20', 'DD/MM/YYYY') from dual;
```

TO_CHAR(DA

20/02/2014

8. TO DATE:

```
Select to_date('23/02/1988', 'DD/MM/YYYY') from DUAL
```

TO_DATE('

23-FEB-88

9. ADD MONTHS:

```
Select add_months(sysdate,4) from dual
```

ADD_MONTH

17-JUN-14

```
Select add_months(date '1994-09-20',4) from dual
```

ADD_MONTH

20-JAN-95

10. LAST DAY

```
Select sysdate, last_day(sysdate) from dual
      SYSDATE LAST_DAY(
```

```
-----
17-FEB-14 28-FEB-14
s
```

11. NEXT DAY

```
Select sysdate, next_day(Sysdate, 'Monday')
from dual
```

```
      SYSDATE  NEXT_DAY(
-----  -----
17-FEB-14    24-FEB-14
```

```
Select sysdate, next_day(date '2014-09-20', 'sunday')
from dual
```

```
      SYSDATE  NEXT_DAY(
-----  -----
20-FEB-14    21-SEP-14
```

```

drop table College;
drop table Student;
drop table Apply;
create table College(cName varchar2(10) primary key, state
varchar2(10), enrollment int);
create table Student(sID int primary key, sName varchar2(10), GPA
real, sizeHS int);
create table Apply(sID int, cName varchar2(10), major varchar2(20),
decision char(1), primary key(sID, major, cName), constraint sID_fk
Foreign key(sID) references Student, constraint cName_fk Foreign
key(cName) references College);

delete from Student;
delete from College;
delete from Apply;

insert into Student values (123, 'Amy', 3.9, 1000);
insert into Student values (234, 'Bob', 3.6, 1500);
insert into Student values (345, 'Craig', 3.5, 500);
insert into Student values (456, 'Doris', 3.9, 1000);
insert into Student values (567, 'Edward', 2.9, 2000);
insert into Student values (678, 'Fay', 3.8, 200);
insert into Student values (789, 'Gary', 3.4, 800);
insert into Student values (987, 'Helen', 3.7, 800);
insert into Student values (876, 'Irene', 3.9, 400);
insert into Student values (765, 'Jay', 2.9, 1500);
insert into Student values (654, 'Amy', 3.9, 1000);
insert into Student values (543, 'Craig', 3.4, 2000);
insert into College values ('Stanford', 'CA', 15000);
insert into College values ('Berkeley', 'CA', 36000);
insert into College values ('MIT', 'MA', 10000); insert
into College values ('Cornell', 'NY', 21000); insert
into College values ('Harvard', 'MA', 50040); insert
into Apply values (123, 'Stanford', 'CS', 'Y'); insert
into Apply values (123, 'Stanford', 'EE', 'N'); insert
into Apply values (123, 'Berkeley', 'CS', 'Y'); insert
into Apply values (123, 'Cornell', 'EE', 'Y');
insert into Apply values (234, 'Berkeley', 'biology', 'N'); insert
into Apply values (345, 'MIT', 'bioengineering', 'Y'); insert into
Apply values (345, 'Cornell', 'bioengineering', 'N'); insert into
Apply values (345, 'Cornell', 'CS', 'Y');
insert into Apply values (345, 'Cornell', 'EE', 'N'); insert
into Apply values (678, 'Stanford', 'history', 'Y'); insert
into Apply values (987, 'Stanford', 'CS', 'Y'); insert into
Apply values (987, 'Berkeley', 'CS', 'Y'); insert into Apply
values (876, 'Stanford', 'CS', 'N'); insert into Apply
values (876, 'MIT', 'biology', 'Y');
insert into Apply values (876, 'MIT', 'marine biology', 'N');
insert into Apply values (765, 'Stanford', 'history', 'Y');
insert into Apply values (765, 'Cornell', 'history', 'N');
insert into Apply values (765, 'Cornell', 'psychology', 'Y');
insert into Apply values (543, 'MIT', 'CS', 'N');
commit;

```


Practical Assignment - 7

Department: Computer Engineering & Applications

Course: BCA

Subject: Database Management System Lab (BCAC0816)

Year: 1ST

Semester: 2ND



Create these tables which consist of following attributes

College

Column Name	Data type	Size
cName	varchar2	10
cstate	varchar2	10
enrollment	int	

Student

Column Name	Data type	Size
sID	int	
sName	varchar2	10
CGPA	number	2,1
marks	int	
DoB	date	

Apply

Column Name	Data type	Size
sID	int	
cName	varchar2	10
major	varchar2	20
decision	char	1

Insert the following data to these tables:

Student

sID	sName	CGPA	Marks
123	Amit	3.9	1000
234	Balbir	3.6	1500
345	chirag	3.5	500
456	Dev	3.9	1000
567	Eshan	2.9	2000
678	Faizal	3.8	200
789	Garvit	3.4	800
987	Himani	3.7	800
876	Ishan	3.9	400
765	Jay	2.9	1500
654	Amit	3.9	1000
543	chirag	3.4	2000

College

cName	state	enrollment
Stanford	CA	15000
Berkeley	CA	36000
MIT	MA	10000
Cornell	NY	21000
Harvard	MA	50040

Apply

sID	cName	major	decision
123	Stanford	CS	Y
123	Stanford	EE	N
123	Berkeley	CS	Y
123	Cornell	EE	Y
234	Berkeley	biology	N
345	MIT	bioengineering	Y
345	Cornell	bioengineering	N
345	Cornell	CS	Y
345	Cornell	EE	N
678	Stanford	history	Y
987	Stanford	CS	Y
987	Berkeley	CS	Y
876	Stanford	CS	N
876	MIT	biology	Y
876	MIT	marine biology	N
765	Stanford	history	Y
765	Cornell	history	N
765	Cornell	psychology	Y
543	MIT	CS	N

State the SQL Queries for each of the following:

- Q1. Create a new column DoB in Student table. (Datatype will be *date*)
- Q2. Insert DoB for each Student in corresponding table using above *instance* of Student table.
- Q3. Find average of CGPA round off to 2 decimal places.
- Q4. Find year of DoB of Student having less than 1000.
- Q5. Display name of *all* Students in uppercase and name of college they applied in lower case.
- Q6. Find fourth alphabet of each student. (Hint: *use* substring)
- Q7. Find sID and sName of student whose sName has string length greater than 3.
- Q8. Find floor, ceiling and truncate (to one decimal place) value of average CGPA.
- Q9. Display details of all students whose sID is *even*.
- Q10. Compute Square Root of 900 and 24^7 .
- Q11. Display sID, sname and DoB in this format 'February 26, 2014' Q15. Convert the text '26/02/2014' to date.
- Q12. Compute on which date is next Saturday and last day of this month?